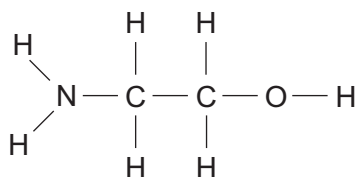


11. (a) 2-Aminoethanol is a colourless liquid.



2-aminoethanol

It is very soluble in water owing to its ability to hydrogen bond with water molecules.

Complete the diagram above to show how 2-aminoethanol is able to hydrogen bond to water molecules. [4]

- (b) The boiling temperatures of 1,2-diaminoethane and 2-aminoethanol are shown in the table.

Compound	Formula	Boiling temperature / °C
1,2-diaminoethane	$\text{H}_2\text{N}-\text{CH}_2-\text{CH}_2-\text{NH}_2$	117
2-aminoethanol	$\text{H}_2\text{N}-\text{CH}_2-\text{CH}_2-\text{OH}$	170

Use these figures to comment on the strength of the intermolecular forces between the molecules in each compound suggesting reasons for your answer. [3]

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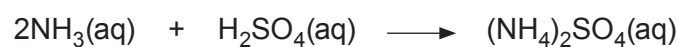
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- (c) Ammonium sulfate can be prepared in the laboratory by neutralising aqueous sulfuric acid with ammonia solution in an acid-base titration.



Outline a method to obtain pure dry crystals of ammonium sulfate by this procedure.

[6 QER]

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11. (a) A solution contains one cation and up to three different anions. The anions possibly present are carbonate, chloride and sulfate.

(i) Devise a plan that unambiguously proves which anions are present in the mixture.

You should also give any observations and conclusions that enable you to identify the anions. [6 QER]

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- (ii) A student said that the cation in the mixture can only be sodium since all the possible anions form a soluble salt with sodium. Is he correct?

Justify your answer and state how you could prove if the statement were true. [2]

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- (b) A student is given four materials and asked to identify the **type of structure** present in each one by carrying out a series of tests.

She is told that the maximum temperature of a Bunsen burner flame is about 800 °C. She is also told that in at least one case, it will **not** be possible to come to a definite conclusion.

Her results are shown in the table below.

	A	B	C	D
Melting temperature / °C	100	>800	>800	>800
Solubility in water	soluble	insoluble	insoluble	soluble
Conductivity of solid	none	none	good	none
Conductivity of solution	none			good

- (i) Use the information in the table to identify each type of structure. Where a definite conclusion cannot be reached explain your reasoning. [4]

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QUESTION CONTINUES ON PAGE 18



- (ii) For **one** of the materials where the type of structure could not be identified, suggest what further test(s) are needed to identify the type of structure. [2]

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- (iii) Suggest why it is difficult to identify a material as a metal when it is in powdered form. [1]

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END OF PAPER



Surname	Centre Number	Candidate Number
Other Names		2

**GCE AS/A LEVEL**

2410U10-1



S19-2410U10-1

CHEMISTRY – AS unit 1**The Language of Chemistry, Structure of Matter and Simple Reactions**

MONDAY, 20 MAY 2019 – MORNING

1 hour 30 minutes

Section A

Section B

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1. to 8.	10	
9.	16	
10.	14	
11.	12	
12.	14	
13.	14	
Total	80	

2410U101
01**ADDITIONAL MATERIALS**

In addition to this examination paper, you will need a:

- calculator;
- **Data Booklet** supplied by WJEC.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Section A Answer **all** questions in the spaces provided.**Section B** Answer **all** questions in the spaces provided.Candidates are advised to allocate their time appropriately between **Section A (10 marks)** and **Section B (70 marks)**.**INFORMATION FOR CANDIDATES**

The number of marks is given in brackets at the end of each question or part-question.

The maximum mark for this paper is 80.

Your answers must be relevant and must make full use of the information given to be awarded full marks for a question.

The assessment of the quality of extended response (QER) will take place in **Q.12(c)**.

If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.



MAY192410U10101

11. A student was asked to find the percentage of calcium carbonate in a sample of chalk. He used the following chemicals.

- Three chalk pieces of identical composition and mass 2.54 g
- Hydrochloric acid solution of concentration 1.00 mol dm^{-3}
- Sodium hydroxide solution of concentration $0.100 \text{ mol dm}^{-3}$

Method

- Use a burette to measure 50.00 cm^3 of hydrochloric acid (an excess) into a 100 cm^3 beaker.
- Put a piece of chalk into the beaker and leave until the reaction finishes.
- Filter the solution into a conical flask to remove any solid impurities.
- Add a few drops of indicator to the solution in the conical flask and titrate against the sodium hydroxide solution.
- Repeat the procedure using the other chalk pieces and calculate a mean titre.
- Use the mean titre to calculate the percentage of calcium carbonate in the chalk sample.

Results

Mass of each chalk piece = 2.54 g

Titration	1	2	3
Final reading / cm^3	16.80	33.05	16.70
Initial reading / cm^3	0.20	16.80	0.35
Titre / cm^3	16.60	16.25	16.35

$$\text{Mean titre} = \frac{16.60 + 16.25 + 16.35}{3} = 16.40 \text{ cm}^3$$

- (a) State how the student would know that the reaction between the chalk and acid had finished. [1]

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(b) Suggest and explain **two** improvements to the student's method. [4]

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(c) The equation for the reaction between calcium carbonate and hydrochloric acid is as follows.



Use this and the student's results, including the mean titre of 16.40 cm^3 , to calculate the percentage of calcium carbonate in the chalk sample. [4]

Percentage = %



(d) Comment on the validity of the mean titre calculated.

[1]

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(e) Another student follows the same procedure but filters the solution before the reaction is complete. State what effect, if any, this would have on the value of the titre. Justify your answer.

[2]

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11. (a) Arsenic oxide, As_2O_3 , is prepared on an industrial scale by roasting arsenic-containing ores such as arsenopyrite, FeAsS , in air. The other products formed are iron(III) oxide and sulfur dioxide.

(i) State the oxidation state of arsenic in As_2O_3 . [1]

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(ii) Give a balanced chemical equation for the industrial production of As_2O_3 from FeAsS . [2]

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- (b) As_2O_3 is moderately soluble in water. 100 cm^3 of a saturated solution at 25°C contains 2.06 g .

When dissolved in water, the oxide reacts to form arsenous acid.



- (i) Calculate the concentration of the arsenous acid, in mol dm^{-3} , in the saturated solution. [3]

concentration of $\text{H}_3\text{AsO}_3 = \dots\dots\dots \text{mol dm}^{-3}$

- (ii) A solution of arsenous acid has a pH of 5.11.

Calculate the hydrogen ion concentration of this solution. [1]

$[\text{H}^+] = \dots\dots\dots \text{mol dm}^{-3}$



- (c) The formula for arsenous acid can be written as $\text{As}(\text{OH})_3$ since it contains three hydroxyl (OH) groups bonded to arsenic.

Suggest the shape around the arsenic atom in $\text{As}(\text{OH})_3$. Justify your answer by using VSEPR theory. [3]

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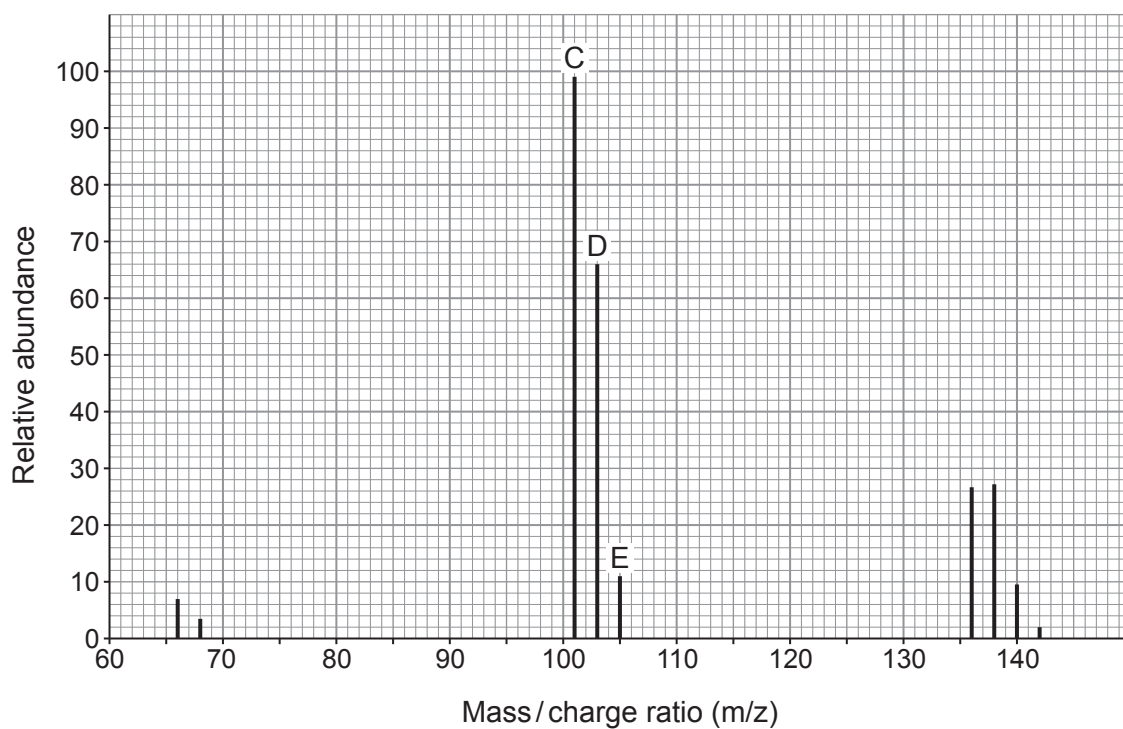
- (d) Phosphorus can form two chlorides, PCl_3 and PCl_5 .

0.181 g of a chloride of phosphorus gave 39 cm^3 of vapour at 1 atm pressure when heated to 87°C . The sample was completely vapourised.

Show that the chloride of phosphorus was PCl_3 . [4]



- (e) The molecular ion region of the mass spectrum of PCl_3 , is shown below.



- (i) Identify the species responsible for peak C at m/z 101.

[1]

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- (ii) Explain why the height ratio of peaks C:E is 9:1.

[2]

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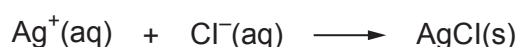


11. Seawater contains a number of dissolved salts. Although composition varies with location, 1 000 cm³ of seawater contains about 20 g of chloride ions, Cl⁻, and about 3 g of sulfate ions, SO₄²⁻.

A student is given a sample of seawater from Rhossili Bay and asked to determine the chloride ion content by volumetric analysis and the sulfate ion content by gravimetric analysis.

- (a) Determination of chloride ion content by volumetric analysis.

The method is similar to an acid-base titration. A silver nitrate solution of known concentration is used to precipitate chloride ions as silver chloride.



The seawater is diluted by a factor of five before it is used in the titration.

The endpoint of this titration is difficult to determine directly, so potassium chromate(VI), K₂CrO₄, is used as an indicator. When all of the chloride ions have been used up, the chromate(VI) ions react with silver ions and produce silver chromate(VI), which forms a red precipitate. The instant a permanent red tinge appears in the solution, the endpoint has been reached.

Volume of diluted seawater in the conical flask = 25.0 cm³

Concentration of silver nitrate solution in the burette = 0.100 mol dm⁻³

Mean titre = 26.40 cm³

- (i) Before starting the titration, the student rinses the burette with silver nitrate solution. Suggest why he does this. [1]

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- (ii) Suggest why the student dilutes the seawater. [1]

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- (iii) Describe how the student should dilute the seawater by a factor of five. [3]

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- (iv) Describe and explain **one** action the student might take just before the endpoint of the titration, to ensure that the volume of silver nitrate added at the endpoint is accurate. [2]

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- (v) Write an ionic equation for the precipitation of silver chromate(VI). [1]

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- (vi) Calculate the mass of chloride ions in $1\,000\text{ cm}^3$ of the original seawater, giving your answer to an **appropriate** number of significant figures. [4]

Mass of chloride ions = g



- (b) Determination of sulfate ion content by gravimetric analysis.

100 cm³ of undiluted seawater and 0.100 mol dm⁻³ barium nitrate solution were used.

The mass of the barium sulfate precipitate was 0.65 g.

You may assume that **all** of the sulfate ions in the seawater were precipitated.

- (i) Describe how the student carried out the gravimetric analysis to find the mass of the barium sulfate precipitated. [5]

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- (ii) Calculate the minimum volume, in cm^3 , of barium nitrate solution needed to precipitate all of the sulfate ions in 100 cm^3 of the seawater.

[3]

Volume = cm^3

- (iii) Suggest why the volume of barium nitrate needed was different to the volume of seawater used.

[1]

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END OF PAPER

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